

Assignment - II

* Dataset:

Windspeed	Blade Angle	Rotor Speed	Power Output
5	11	125	48
7	9	155	72
9	15	175	101
11	13	220	142
13	18	285	176

ML Model Equation.

$$* \{ Y = M_1 X_1 + M_2 X_2 + M_3 X_3 + C \}$$

$$\therefore X = \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 155 \\ 1 & 9 & 15 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 285 \end{bmatrix}$$

$$Y = \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 11 & 9 & 15 & 18 & 18 \\ 125 & 155 & 175 & 220 & 285 \end{bmatrix}$$

$$X^T X = X^T \times X \Rightarrow \begin{bmatrix} 5 & 45 & 66 & 910 \\ 45 & 445 & 630 & 8760 \\ 66 & 630 & 910 & 12485 \\ 910 & 8760 & 12485 & 173900 \end{bmatrix}$$

$$X^T Y = X^T \times Y \Rightarrow \begin{bmatrix} 539 \\ 5503 \\ 7705 \\ 107435 \end{bmatrix}$$

$$\therefore \theta = (X^T X)^{-1} X^T Y$$

$$\theta = (X^T X)^{-1} X^T Y$$

$$\theta = \begin{bmatrix} -99.911 \\ 1.217 \\ 2.417 \\ 0.906 \end{bmatrix} = \begin{bmatrix} C \\ M_1 \\ M_2 \\ M_3 \end{bmatrix}$$

$$\therefore \left\{ Y = 1.217x_1 + 2.417x_2 + 0.906x_3 - 99.911 \right\} \Rightarrow \underline{\text{ML Model.}}$$

For, $x_1 = 9$

$x_2 = 11$

$x_3 = 100$

$$Y = 1.217 \times 9 + 2.417 \times 11 + 0.906 \times 100 - 99.911$$

$$Y = 10.953 + 26.587 + 90.6 - 99.911$$

$$\underline{\underline{Y = 28.229 \text{ kW}}}$$

$\therefore$ *Predicted power output = 28.229 kW

* Intercept - $c = -99.911$

* Windspeed coefficient = $m_1 = 1.217$

* Blade angle coefficient = $m_2 = 2.417$

* Rotor speed coefficient = $m_3 = 0.906$